

BUILDING MATERIALS

‘Solutions strategy’, a driver of outperformance, or a way to broaden the portfolio?

Over the last 5 years, solutions selling has become the key medium term strategy for several building materials stocks, with mentions of “building solutions” or “building envelope” mentioned 3x more on conference calls.

One strategy doesn’t fit all: The rationale is often to see greater organic growth through increasing the market share of their existing products, or to increase the scale of their portfolio through inorganic opportunities (Net acquisitions in US\$ are 2x higher the last 5 years). We believe solutions strategy can be a differentiator in taking share and growing profitability for some categories, but do not believe it is a universal approach all companies can benefit from, particularly in more commoditised or unrelated products.

Stock implications: Within our coverage we see **Kingspan/Sika** as having the strongest strategies to drive outperformance vs. end-markets, with upside to our forecasts for **St. Gobain/Holcim** if they can execute on bundling their large diversified portfolios. However, we expect companies like Holcim to continue pursuing the strategy (inline with 2030 M&A envelopes) given it provides an accretive use of capital and diversification benefits away from the core business.

What is ‘Solutions Selling’? Solutions selling looks to package complementary products from a single manufacturer to serve a defined construction problem vs. providing individual components. For example, the strategy is common in roofing (which needs to be weather-resistant and long-lasting), stormwater drainage (used within public infrastructure such as highways, urban areas) and basements/tunnels (need to be structurally sound and moisture resistant).

What are effective solutions strategies? We believe there are 3 dimensions of an effective solution strategy: **(1)** Product offering – selling interrelated products in systems with warranties and reliant on specific high performance products, **(2)** Targeted distribution – i.e. selling into the same decision maker and helping resolve pain points, and **(3)** Margin capture – bundling in high margin complementary products.

Ben Rada Martin
+44(20)7051-0800 |
ben.radamartin@gs.com
Goldman Sachs International

Susan Maklari
+1(212)357-3906 |
susan.maklari@gs.com
Goldman Sachs & Co. LLC

Charles Perron-Piche, CFA
+1(917)343-1847 | charles.perron-
piche@gs.com
Goldman Sachs & Co. LLC

Natasha Phillips
+44(20)7051-4705 |
natasha.phillips@gs.com
Goldman Sachs International

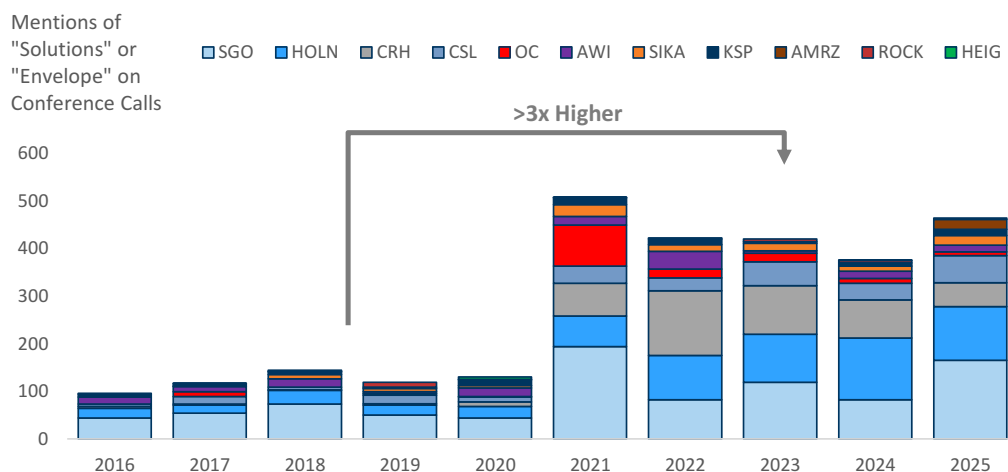
Patrick Creuset
+33(1)4212-1380 |
patrick.creuset@gs.com
Goldman Sachs Bank Europe SE – Paris
Branch

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Which solutions strategies are genuine growth engines?

In the last 5 years the “solution” or “envelope” strategy from large Building Materials companies, with the terms mentioned 3x more often in the last 5 years than the prior 5 in conference calls.

Exhibit 1: Solutions strategy has been in vogue the last few years as companies look to push synergies between products or expand their portfolio

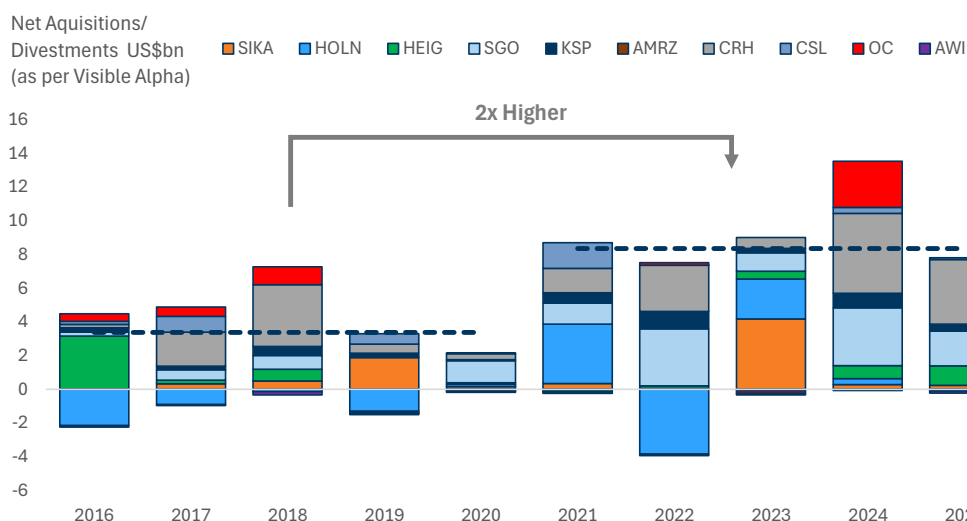


Source: Bloomberg

Inorganic M&A activity has picked up

We believe solutions strategies can become a driver of outperformance in specific categories, for others is broadly a way to deploy capital and expand the portfolio, hence unlikely to drive financial performance. We note net acquisitions (acquisitions - divestments) has doubled on a nominal US\$ basis in the last 5 years vs. the prior 5.

Exhibit 2: M&A activity in the sector has coincided with the pickup in solutions/envelope strategies



Source: Visible Alpha Consensus Data

The strategies we are most positive on in driving margin uplift and excess volume growth include:

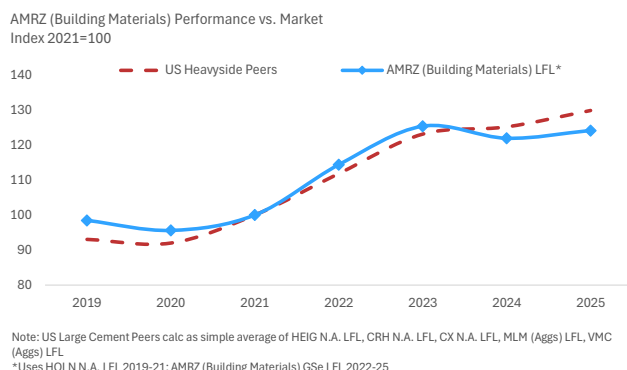
- **Sika** - Sika's long term solutions focus across various categories (roofing, basements, infra projects) and category leading R&D spend are key strategies for outgrowing endmarkets its peers ([see prior analysis](#)). We believe Sika has a solid product advantage within solutions selling, supported by higher R&D spend vs. peers, and is able to leverage their warranties to create demand pull-through. For example, Sika's newer basement waterproofing SikaProof A+ membrane has seen +27% sales CAGR FY'23-FY'25 and offers a 20-year warranty conditional on dual use with Sika's concrete admixtures. Sika's solutions selling effectively targets decision-makers at long lead times in the construction process and offers complimentary products to solve defined construction problems. For instance, structural engineers requiring groundwater-protected below-ground structures and architects/building owners for fire-resistant roofing systems (i.e. Sika's PVC roofing). We believe Sika's high-value solutions and leading R&D results in material margin capture noting 23-25% of sales are from products <5 years old (typically capture +3-5pp higher margin). We see Sika as driving outperformance vs. peers.
- **Kingspan** - We believe Kingspan's US roofing expansion - which looks to extend existing strength in the panels market (GSe 40% share) into the high margin non-resi roofing market - will provide a differentiated full exterior product solution (roofing + insulated wall panels) with up to a c.40-year warranty. We see potential to cross-sell between exterior/roofing solutions given overlap in key decision-makers (i.e. project manager, building owner) and exterior solutions offering an earlier lead time vs. roofing. Yet, we remain cautious on ramp-up given roofing incumbents benefit from contractor loyalty and strong distributor relationships which we see as barriers to overcome. We continue to monitor the situation as it develops, noting initial Polyiso production has started whilst TPO membrane is expected by 4Q'26.

Broader solutions strategies also remain a focus for:

- **Holcim** - recently completed its acquisition of European walling provider Xella, the first major push into solutions strategy for Holcim (given Amrize business pre-spin was key M&A focus). For Xella, we see margin capture through cost synergies, and EPS accretion as key positives of the solutions M&A push, with upside to our forecasts if revenue synergies can be achieved. The reason why we remain conservative on revenue opportunities, is that we believe concrete remains relatively commoditised (customers are price sensitive, minor specification selling), with different decision makers (architect vs. foreman), and limited system benefits between foundations and walling. We note **AMRZ**, HOLN's US spin-off ([Jun-25](#)), pursued a similar strategy to provide the advanced solutions for the building envelope (acquisitions in US Non-resi + Resi roofing), however has seen both its core materials business track inline with heavyside peers and its envelopes business inline with category peers since.

Exhibit 3: AMRZ solutions strategy has performed inline on topline vs. broader heavyside peers since its 2022 building envelope push...

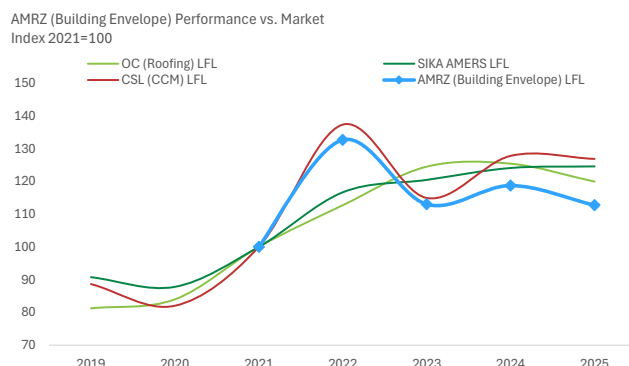
AMRZ (Building Materials) Performance vs. Market, Index 2022=100



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 4: ...and roofing peers

AMRZ (Building Envelope) Performance vs. Market, Index 2021=100

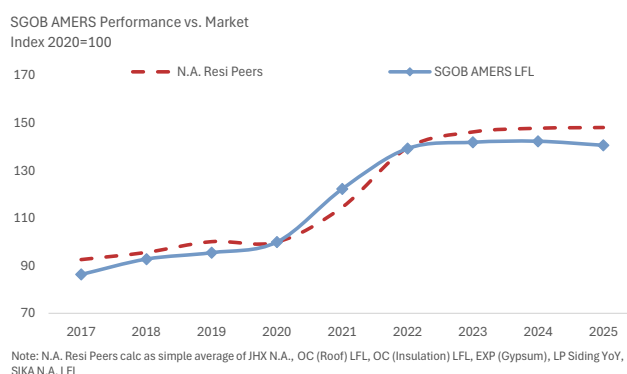


Source: Company data, Visible Alpha Consensus Data, Goldman Sachs Global Investment Research

- **Saint Gobain** - is pursuing a full building envelope solutions strategy across lightside products (gypsum, glass, roofing, siding, construction chemicals). SGO sees upselling as a meaningful driver of volume share and profitability for - with an illustrative >15x profit / >10x revenue uplift presented at the Oct-25 CMD. We note this strategy was also used by SGO's Australian subsidiary CSR, with limited evidence of outperformance. Furthermore, given the relative commoditisation of these products, we are yet to see outperformance in North America (where SGO's solutions offering is the strongest) vs local peers.

Exhibit 5: SGOB has seen limited topline outperformance vs. N.A. peers

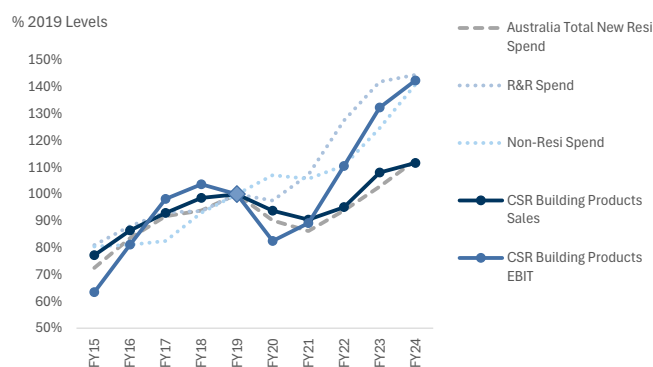
SGOB AMERS Performance vs. N.A. Resi Peers (Index 2020=100)



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 6: CSR's revenue has largely performed in line with Australian New Resi spend...

CSR Standalone Revenues / Australia Nominal Construction Spend - % 2019 levels (Mar y/e)



Source: ABS, Company data

Other names in our coverage (Heidelberg, Rockwool and Buzzi) are not actively pursuing a solutions strategy. We note that small acquisitions from **Heidelberg** are in downstream areas that could be complementary to a solutions strategy - see recently announced minority investment in AmeriTex Pipe & Products ([May-26](#)).

Global building materials peers are also implementing solutions strategies with strong

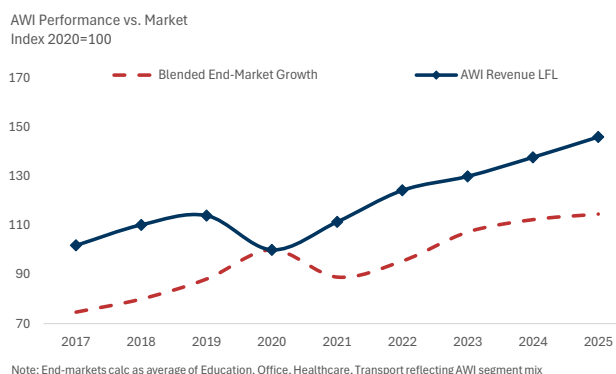
success including: **(1) Armstrong (covered by Susan Maklari):** The launch of innovative new products along with the integration of M&A in architectural specialties has driven market relative outperformance over the last several years. The growth of the AS platform is also expected to drive volumes of higher margin mineral fiber products as the company gains exposure to more complex projects.

(2) Carlisle (covered by Susan Maklari): has the largest market share in North American Flat roofing solutions underpinned by an increased focus on innovation, including a gradual ramp in R&D spend to 3% of sales by 2030 vs <1% in 2023 supporting innovative market solutions with 10+ new products set to launch in 2026 (16 ft TPO membrane), effectively solving pain points of their loyal contractor base and a trusted warranty programme with the business also outgrowing North American Non-Resi peers since 2018.

(3) CRH (not covered) is pursuing a solutions strategy by cross-selling its core cement and aggregates business into growing public infrastructure end-markets (water, highways, energy, reindustrialisation). CRH has been building out their solutions business via acquisitions which build out their solutions' expertise and portfolio, with the recently proposed acquisitions of Arcosa (Jun-26) in energy infra and Axius (May-26) in water quality.

Exhibit 7: AWI has outperformed its blended end-market growth since 2020...

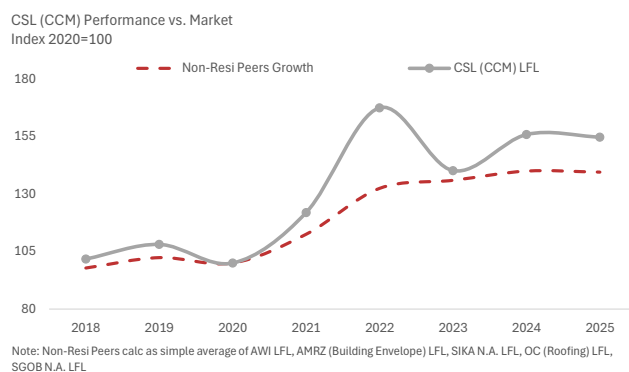
AWI Performance vs. Blended End-Market Growth



Source: Eikon Datastream, Company data, Goldman Sachs Global Investment Research

Exhibit 8: CSL Flat roofing systems has outperformed North American non-resi peers...

CSL (CCM) Performance vs. N.A. Non-Resi Peers (Index 2020=100)



Source: Company data, Goldman Sachs Global Investment Research

We believe 3 dimensions drive an effective solutions strategy:

- 1. Product advantages:** can generate a more defensible pricing premium and capture greater TAM by offering: **(A)** Systems solution for a defined construction problem (i.e. basement waterproofing); **(B)** Warranties covering technically complex, multi-component systems where faults can result in costly repair or structural damage; and **(C)** Anchor products that provide a technological advantage (i.e. Panels) or certain technical expertise (water infra regulation), noting these also generate pull-through demand.
- 2. Sales network connectivity:** provide volume upside for cross-selling opportunities via: **(A)** Longer lead times on high specification products, in combination with **(B)** Single vendor relationships support up-selling of adjacent products by removing

friction and combining purchasing decisions. This can be particularly effective for upselling high-volume, low-margin products such as cement via complementary services (infra construction expertise).

3. **Margin capture:** can be driven by products that are in high margin adjacencies or are verticalised in their cost structure.

Exhibit 9: We outline our framework against existing solutions strategies within building materials

GS Framework for Understanding Solutions Strategies

	Product Bundling	Sales Network	Margin
SIKA Basement Solutions	✓✓✓	✓✓✓	✓✓
KSP + Panels/Roofing Solutions	✓✓	✓	✓✓
SGOB Exterior Solutions	✓✓		✓
HOLN Solutions	✓	✓	✓✓
CRH + Civil Infra (water, energy)	✓✓	✓✓	✓✓
Interior Solutions	✓✓✓	✓✓	✓✓✓
Non-Resi Roofing Systems	✓✓✓	✓✓	✓✓

Source: Company data, Goldman Sachs Global Investment Research

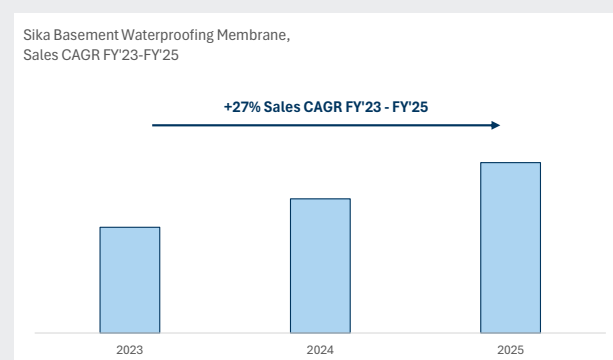
SIKA Basement Solutions Strategy

Within our coverage, we see **SIKA**'s basement products as a unique example of effective solutions selling. We believe below-ground waterproofing presents a **(1)** specialised product with strong warranty; **(2)** early entry and continued involvement in the construction process; and **(3)** solid profitability. Our key views include:

- **Product bundling** - SIKA leverages their expertise in thermoplastic membranes, mortars and concrete admixtures to provide full basement systems solutions addressing water, gas, biological and temperature variation pressure to protect the building from structural deterioration. Sika offers 15-year (single-system watertight concrete admixtures) and 20-year (dual-system, with additional waterproof membrane) guarantees on products.
- **Sales Channel** - Project owners and structural engineers are heavily invested in the design and quality of below-ground waterproofing, providing SIKA early access into the construction process. SIKA also offers seminars educating key stakeholders on applications, sustainability and varying designs creating an early touch point with decision-makers. Despite waterproofing systems typically <1% of total building cost, quality design and installation significantly enhances the value proposition to project owners via savings on future maintenance & repair costs over the structure's lifespan.
- **Margin capture** - Sika's patented waterproofing membrane system grew sales +27% CAGR between FY'23-FY'25. We believe their high-value solutions are driven by a category-leading R&D engine which supports 23-25% of sales from products <5 years old, typically resulting in +3-5pp higher material margins.

Exhibit 10: Highly specified, differentiated and innovative product offerings can drive meaningful sales performance...

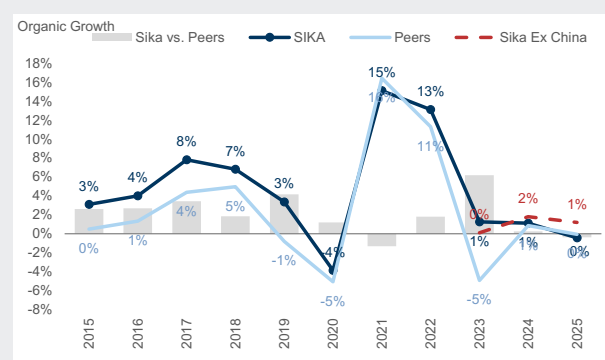
Sika Basement Waterproofing membrane, Sales Growth



Source: Company data

Exhibit 11: ...Sika's R&D engine has historically supported outperformance vs. peers

Sika Organic Growth vs Peers



Source: Company data, Goldman Sachs Global Investment Research

Dimension 1: Product Advantages

We believe certain products that: **(A)** can be sold in systems; **(B)** where customers value guarantees (warranty); **(C)** where a product is specified in the design; will take share from other products and attract more defensible pricing premiums.

A) Systems Selling

What's is 'Systems selling'? Systems selling is the sale of a bundle of products that are co-reliant on each other for function.

How does it add value to customers? It improves the functionality of the construction product, by ensuring that the multiple components work with each other, and also allows ease of access when it comes to procurement (ordering, delivery, installation).

How does it provide the potential for revenue growth for building materials companies? It allows the manufacturers to capture a greater portion of the bill of materials and in some cases charge higher ASP to reflect the greater production functionality or guarantees.

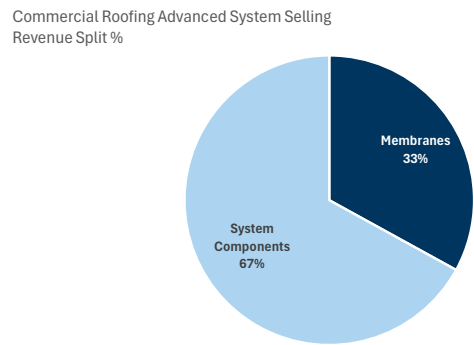
Systems Selling Strategies

- **Flat roofing system (Carlisle, Sika, Kingspan):** provides the individual layers of a flat roofing system, including the vapour barrier, insulation, components (fasteners) and the core membrane. These allow for guarantees around the full system vs individual components (see warranty section). Players that provide this offering include Sika, Carlisle, Amrize, and Kingspan.
- **Below-ground systems (Sika):** Similarly, Sika has developed a suite of products for below-ground systems solutions, these include waterproofing membranes, sealants, cementitious waterproofing mortars and bitumen-based coatings. These products are highly technical and designed to withstand groundwater ingress and contact in basements and tunnels.
- **Exterior system (Kingspan):** Kingspan has a solid exterior panels business which uses its core insulation offering to which it is looking to expand into US roofing systems (currently building-out capacity). This will allow Kingspan to offer full exterior shell solutions.
- **Public infrastructure (CRH):** CRH leverages its public infrastructure platform which serves water, energy and highway end-markets to bundle services with its cement/aggs business, noting >80% of products within the water segment consuming aggregates and cement.
- **Other systems selling:** includes **(1)** Saint Gobain's exterior solutions (combining glass, siding, waterproofing, insulation, gypsum); **(2)** Holcim envelope strategy providing both walling (Xella), foundations (core Holcim business) and roofing (e.g. Elevate/Zinco); **(3)** Amrize's building solutions strategy leveraging strong positions in Non-resi roofing and its building materials portfolio (concrete also used in non-resi roofing); and **(4)** Resi roofing systems (SGO, OC) similar to non-resi component/roof cross selling.

GS View: Overall, we believe certain categories that require a greater product assurance (e.g. Roofing/Basement systems) provide the highest value when sold as a 'system'. These projects require high levels of technical complexity across multi-layered products

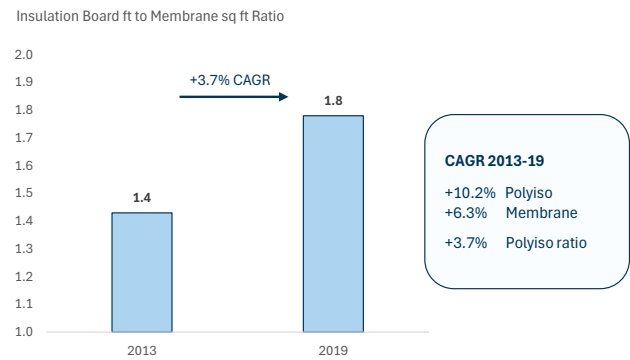
which, alongside a single seller guarantee, can become a high-value, high-utility offering. In contrast, exterior and interior solutions are constructed with more standardised materials where defects are less meaningful, are more visible and easier to fix.

Exhibit 12: Solutions selling expands revenue opportunities into adjacent products...
% Revenue Split in Flat Roofing Systems



Source: Company data

Exhibit 13: ...with opportunities to upsell core materials i.e. Polyiso market grew 1.6x CAGR Membranes 2013-19
Carlisle - ratio of Insulation Board (ft) to Membranes (sq ft)



Source: Company data

B) Warranty

Warranties are guarantees provided by the manufacturer/seller on product performance and lifespan, repairs and failures required during the warranty period are covered by the provider.

Warranty Strategies

- **Non-resi roofing warranties (Carlisle, Kingspan, Sika, Amrize):** highly valued by building owners given high risk of structural or interior impairment from weather-related damage. Typically cover c. 30-years with variations in coverage and requirements for manufacturer, installers, and repairs to be covered. For example, Sika warranties are guaranteed only with Sika's certified 1,400 contractors.
- **Basement warranties (Sika):** Similar to roofing, faults or damage due to poor design or installation has a disproportionate consequence on the structural integrity of the building and require costly repair/refitting. Hence, warranties carry meaningful weight within the value proposition of a basement solution. Sika offers 20-year guarantee on specified dual-system waterproof membranes and concrete admixtures when installed by a Sika-trained contractor.
- **Exterior systems (Kingspan):** Kingspan provides <40-year thermal warranty for their insulated Quadcore panels, with potential of a full single-system warranty when combined to their roofing systems (full exterior offering), subject to review and adherence to design and installation principles outlined in the product details.

How does it add value to customers? High performance building products typically require strong expertise in development and installation to function correctly. Faults in production or improper fitting can result in a suboptimal product performance necessitating further repair or replacement cost. If the fault is with the manufacturer, warranties provide assurance that, conditional on pre-specified requirements the cost will not be at the expense of the project owner.

How does it provide potential for revenue growth for building materials companies?

Project owners are willing to pay greater cost upfront to minimise the future cost of repairs and maintenance during the building lifecycle.

GS view: We believe warranties can be a key aspect of the decision-making criteria for solutions strategies as they reduce potential future spend of the product owner. We note single-vendor warranties are particularly attractive as they eliminate ambiguity over which manufacturer bears the risk, hence resulting in stickier customer bases. This is particularly effective in roofing and basements solutions where building owners highly prioritise efficient repairs and it is difficult to attribute fault to a single component.

C) Specialised Product Offering

What is a Specialised Product Offering? Products or services that provide a meaningful technological advantage or greater expertise vs. peers which underlies higher market share or pricing premium.

How does it add value to customers? Specialised product/service offerings with meaningful technical differentiation or human expertise reduce the purchasing and execution risk for customers and delivers superior product performance.

Unique Product Offering Strategies

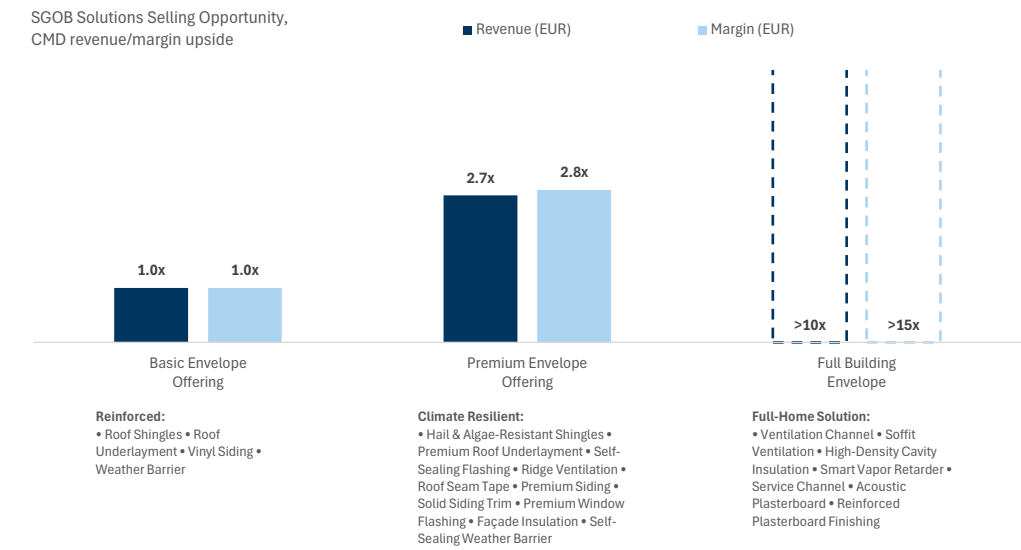
- **Exterior solutions (Saint Gobain, Armstrong):** Saint Gobain offers the complete Resi building envelope to improve their pricing power and create margin uplift. Saint Gobain sees their strategy as an opportunity to capture greater share of wallet (cross-selling), drive high value-add products (up-sell) and create sticky demand early in the design process (specification-driven sales). Armstrong designs and manufactures exterior architectural applications for non-resi buildings. Armstrong is a trusted brand for their expertise in creating innovative and highly specified solutions for architects and designers.
- **Membranes (Sika, Carlisle):** Sika offers high-quality, multi-functional (i.e. waterproof, sound- and fire-resistant, self healing) membranes in conjunction with its adhesives ensuring a strong bond between membrane and substrate that prevents water seeping through the joints. Carlisle developed a 16 ft TPO membrane for roofing systems which became the widest TPO sheet on the market creating meaningful labour savings and improving installation efficiency for project owners.
- **Insulation (Kingspan):** Kingspan Quadcore offers superior technical performance relative to foam board peers (i.e. PIR, PUR), including fire protection, thermal efficiency and a 30-year thermal warranty, this is used in projects that require thermal performance (cool rooms, Tesla gigafactories).
- **Infrastructure (CRH):** CRH's water infrastructure platform, Oldcastle Infrastructure, uses a one-stop shop approach to deliver a differentiated value proposition to customers, offering services across materials procurement, regulatory compliance knowledge and design expertise.
- **Other offerings:** includes Xella cement-based (AAC) walling blocks offer an alternative to traditional clay bricks for buildings with greater energy-efficiency, more lightweight and stronger sound/thermal insulation.

How does it provide potential for revenue growth for building materials companies?

Product differentiation shifts decision-making from price to performance, providing companies with defensible pricing power and stickier customer relationships.

GS view: We believe unique products or services offer a platform through which to leverage cross-selling opportunities of more standardised or commoditised products given the customer base is relatively inelastic.

Exhibit 14: ...SGOB sees >15x margin opportunities for verticalising their lightside offering into the Full Building Envelope solution
SGOB estimates on solutions selling opportunity from CMD



Source: Company data

Dimension 2: Sales Network

A) Lead Times

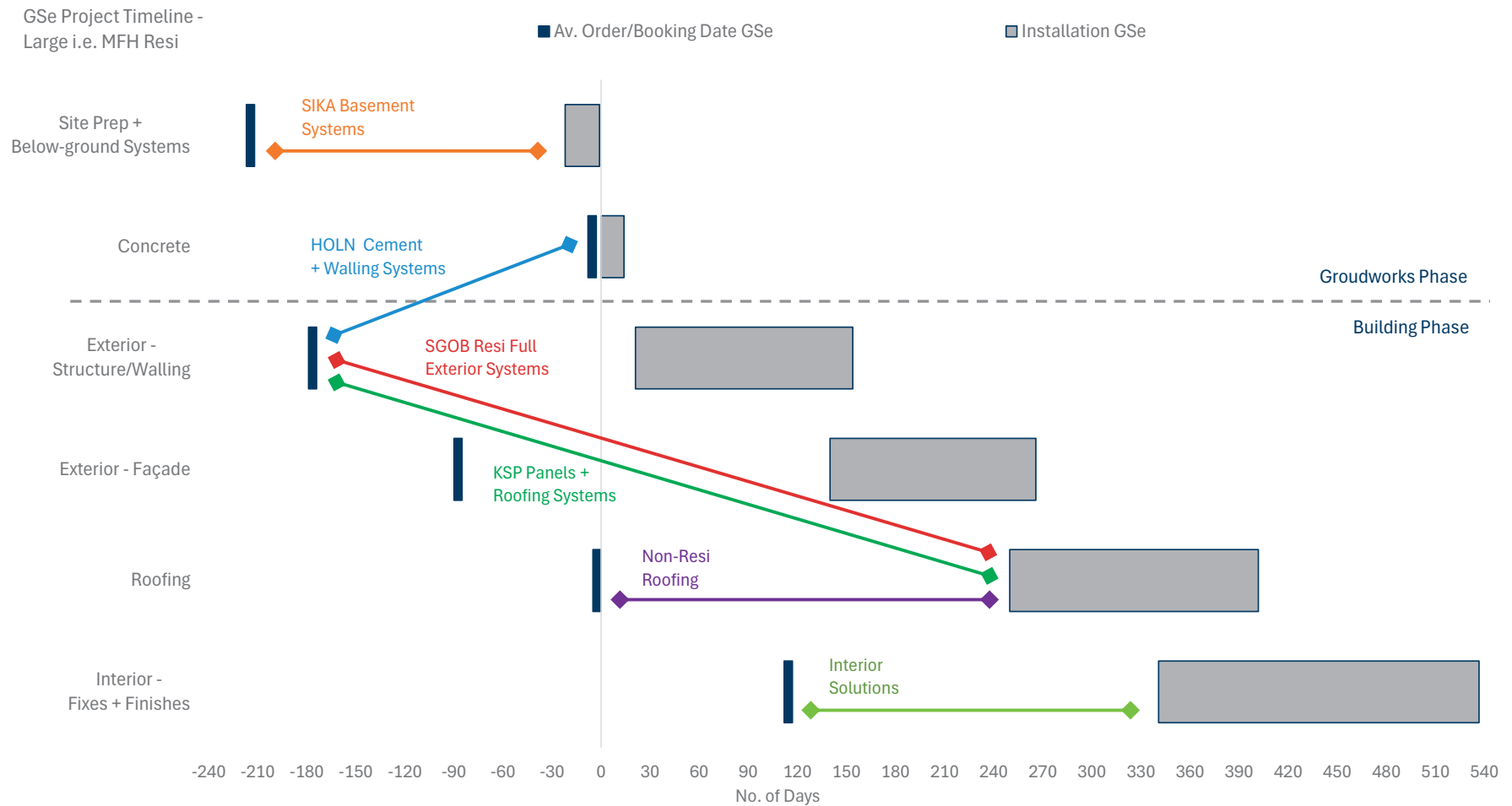
How does it provide potential for revenue growth for building materials companies?

Key design products (i.e. Resi exterior walling structures, architectural exterior facades) are involved in the design process earlier and hence have longer lead times than more standardised materials. Products with greater lead time provide an opportunity for building materials companies to upsell adjacent products, hence capturing larger share of wallet from the project owner.

Lead Times Strategies

- **Public Infrastructure (CRH):** CRH offers precast concrete products such as concrete pipe (for stormwater, wastewater). In civil and state infra projects where large-scale, precast concrete piping is specified early by civil engineers. Once locked in, the pipe solutions act as a pull-through for the bulk materials (i.e. cement/aggs) given overlapping timelines.
- **Exterior/Interior Solutions (Armstrong):** Armstrong leverages its highly specified architectural segment to provide an early lead time to upsell its ceiling product.
- **Other lead time strategies (Kingspan, Amrize, Saint Gobain, Holcim):** Kingspan is aiming to capture greater share of wallet by leveraging its specified insulated panels to pull-through demand for roofing systems. Amrize and Saint Gobain follow a similar strategy within resi/non-resi building solutions to push the full building envelope by entering the project journey during the exterior specification stage. Holcim believes entering the construction process via the exterior spec stage will enable them to pull-through their cement as the project's foundations.

GS view: Complementary product offerings with different lead times can lock customers into a single supplier relationship, enabling greater share of wallet capture for the manufacturer. In our view suitable verticalisation opportunities are split between project phases, with the pre-construction (groundworks, cement pour) phase offering stronger verticalisation opportunities for heavyside whilst construction (exterior, roofing, facade, interior) provides more compelling potential for lightside.

Exhibit 15: Illustrative GSe Timeline for Large Projects (i.e. MFH Resi)

Source: Goldman Sachs Global Investment Research

B) Decision Maker

How does it add value to customers? Construction processes are highly complex given competing and overlapping timelines for product procurement, delivery and installation as well as labour logistics, compliance with local building codes, budgeting and communication between stakeholders. Opportunities for key authorities to streamline decisions and processes prevents delays, budget overruns and potential labour/supply shortages are valued.

How does it provide potential for revenue growth for building materials companies?

Becoming a well-known and trusted partner on construction projects is a highly attractive proposition to the project owner which defends pricing power and allows greater opportunity to up-sell high-volume, low-margin products to capture greater share of wallet.

Decision Maker Strategies

- **Public Infra (CRH):** CRH embeds themselves with their customers (i.e. civil engineers and government officials) by providing full infra project services (design, compliance, manufacturing and on-site) at a single touchpoint. Hence, streamlining decision-making and management for the customer by serving these function in-house.
- **Below-ground structures i.e. basements, parking (Sika)** - Sika builds solutions around solving defined problems for key decision-makers. For example, structural engineers are a critical part of the decision-making process for below-ground solutions and are responsible for minimising the risk of groundwater damage, hence packaging products to prevent leakage via membranes and waterproofing admixtures (cement) is highly effective and creates pull-through demand.
- **Exterior solutions (Kingspan, Amrize, Saint Gobain, Holcim, Armstrong):** Companies focused on exterior solutions tend to target early entry with the architect/project owner during the design and technical specification stage to secure their anchor products, then leverage this position to pull-through core or adjacent offerings.

GS view: We believe the channel partners with the highest leverage are the individuals responsible for the project's strategic vision such as key stakeholders, architect, or civil engineer, given their role includes budgeting, resourcing and project viability. Decision-making responsibilities tends to narrow to specific materials further down the chain-of-command. In conjunction with this, design authority (dictates technical specifications) typically carries more weight vs execution authority (dictates daily methods and operations).

Exhibit 16: Design decisions typically outweigh execution decision-making authority, with Non-Resi and Infra projects giving more design authority on materials
GSe Framework of Project Hierarchies

Stakeholder	Solutions Selling Opportunity	Resi	Non-Resi	Infra
Owner/Key Stakeholder				
Project Manager/Architect				
Site Manager/Lead Engineer				
Specialised Engineer				
Foreman				
Skilled traders (plumbers, etc)				

Source: Goldman Sachs Global Investment Research

Exhibit 17: ...stakeholder involvement varies across phase and components
GSe Framework of Sales Channels across Stakeholders

	Pre-Construction		Construction Phase		
	Groundwork	Concrete	Exterior	Roofing	Interior
Owner			✓✓✓	✓✓✓	✓✓✓
Project Manager/Architect	✓	✓	✓✓✓	✓✓✓	✓✓✓
Site Manager	✓✓	✓✓	✓✓	✓✓	✓✓
Specialised Engineer	✓✓	✓✓	✓	✓	✓
Foreman	✓	✓			
Skilled traders			✓	✓	✓

Source: Goldman Sachs Global Investment Research

Dimension 3: Margin capture

What allows for margin capture? Upside to unit economics can take place when the solutions product sees a high margin from an adjacent category, or a product with verticalised inputs.

Margin Capture Strategies

- **Materials spend in Infra (CRH):** CRH's services across cement/aggs, highways and water infra provide meaningful supply-chain synergies, noting >80% of products within the water segment consume aggs/cement as well as >85% of roads require water management systems. This also creates synergies in their sales and distribution network as end-customers and job sites commonly overlap, CRH reported c. 30% EBITDA FY'25 margin on their water infra business (USD1.6bn revenue, USD500m EBITDA).
- **High margin, low ticket price products (Armstrong):** Armstrong's Architectural Specialties segment reported adj. EBITDA margin of 15.3% in FY'25, whilst their Mineral Fiber segment reported 42.1% adj. EBITDA FY'25. Armstrong leverages their high specification architectural exterior/interior products to cross-sell their high volume, low price ceiling tiles as a packaged service streamlining decision-making for architects on interior solutions.
- **Walling Solutions (Holcim):** Holcim believes they will see EUR 60m in synergies for the Xella acquisition, with c.1/2 of these from costs as Xella benefits from raw materials procurement synergies (AAC blocks are reliant on cement, lime and fly ash). Hence we believe this makes the product somewhat lucrative for Holcim to push given its verticalisation of the cost base.

How does it add value to the customer? Greater product profitability can be passed on by the manufacturers in more competitive pricing.

How does it provide potential for revenue growth for building materials companies? Greater profitability allows for the company to be more competitive on price or to capture more of the unit economics.

GS view: In our view, we expect the pull through of high margin adjacent products is a selective opportunity for some smaller product categories. Nevertheless it can be a meaningful driver of outsized earnings growth in these cases.

Valuation Methodologies & Key Risks

Armstrong World Industries Inc. (Buy, covered by Susan Maklari): Our 12-month price target of \$210 is based on 13.8x our Q5-Q8 adj. EBITDA estimate. **Downside risks include: 1)** increase in raw material costs and the inability to pass them through, **2)** delays or the inability to deliver on strategic initiatives, **3)** new products fail to gain traction at expected pace, **4)** inability to integrate recent acquisitions, and **5)** changes in WAVE joint venture could have an adverse impact on the company.

Holcim (Buy): Our 12m price target of CHF 88 is based on an equally weighted DCF valuation of CHF 88/share (WACC 7.2%, TGR +1%) and a PE valuation of CHF 87/share.

Key downside risks to our view and price target: (1) More significant and sustained energy inflation driving operating cost pressure; **(2)** A weaker-than-expected Europe cycle driven by German fiscal and new resi recovery; **(3)** European peer rationality pushing volume at the expense of price; **(4)** Dilutive M&A in the solutions business; **(5)** CBAM enforcement and timeline of phase-outs impacting relative outperformance; **(6)** Latam/Mexico margin sustainability against competition; **(7)** CCUS bottlenecks from transport and storage partnerships; and **(8)** Decarbonisation and CCUS capex overruns.

Kingspan (Neutral): Our 12-month price target of €80 is based on a P/E multiple of 18x applied to our forward estimates (weighted 75% FY27E, 25% FY28E EPS). **Upside risks: (1)** Commercial traction in US Roofing as KSP looks to launch in 2H'26; **(2)** Greater DC demand duration; **(3)** New product growth drivers in Panels; **(4)** Further European Insulated Panels/Foam consolidation. **Downside risks: (1)** Delays in US Roofing ramp; **(2)** Negative price/cost from energy inflation.

Saint-Gobain (Neutral): Our 12-month price target of €84 is based on a P/E multiple of 12x applied to our forward estimates (weighted 75% FY27E EPS, 25% FY28E). **Upside risks to our view and price target: (1)** Stronger-than-expected European construction cycle; **(2)** Continued portfolio rotation of lower-margin businesses; **(3)** Greater evidence on structural growth drivers; **(4)** Additional cost out programs; **(5)** Greater-than-expected shareholder returns. **Downside risks to our view and price target: (1)** Higher-for-longer interest rates as a key driver of the construction cycle; **(2)** Lower consumer confidence; **(3)** Weaker-than-expected operating leverage; **(4)** Dilutive M&A; **(5)** Greater pricing competition or new volume capacity from competitors.

Sika (Buy): Our 12-month TP, based on a 12mf price/earnings methodology (24x multiple), is CHF 210/sh. **Downside risks: (1)** Weaker-than-expected construction cycle in the US and China given Sika's exposure; **(2)** Greater-than-expected energy cost inflation with political instability and tariffs; **(3)** Slower forward cadence of bolt-on M&A; **(4)** Limited revenue impact from new products; **(5)** Elevated competition given the financial attractiveness of the construction chemicals category; **(6)** Limited forward operating efficiencies as the company matures.

Disclosure Appendix

Reg AC

We, Ben Rada Martin, Susan Maklari, Charles Perron-Piche, CFA, Natasha Phillips and Patrick Creuset, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Ben Rada Martin Goldman Sachs International, Susan Maklari Goldman Sachs & Co. LLC, Charles Perron-Piche, CFA Goldman Sachs & Co. LLC, Natasha Phillips Goldman Sachs International, Patrick Creuset Goldman Sachs Bank Europe SE – Paris Branch.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% – Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

Rating and pricing information

Armstrong World Industries Inc. (Buy, \$158.56), Holcim (Buy, SFr74.88), Kingspan (Neutral, €79.20), Saint-Gobain (Neutral, €79.86) and Sika (Buy, SFr172.70).

The rating(s) for Holcim, Kingspan, Saint-Gobain and Sika is/are relative to the other companies in its/their coverage universe: Buzzi, Heidelberg Materials, Holcim, Kingspan, Rockwool, Saint-Gobain, Sika

The rating(s) for Armstrong World Industries Inc. is/are relative to the other companies in its/their coverage universe: A.O. Smith Corp., Alliance Laundry Holdings Inc., Armstrong World Industries Inc., Boise Cascade Co., Builders FirstSource Inc., CSW Industrials Inc., Carlisle Companies Inc., D.R. Horton Inc., Fortune Brands Innovations Inc., Installed Building Products Inc., JELD-WEN Holding, KB Home, Latham Group Inc., Lennar Corp., Louisiana-Pacific Corp., Masco Corp., Meritage Homes Corp., Mohawk Industries Inc., Owens Corning, Pool Corp., PulteGroup Inc., SiteOne Landscape Supply Inc., Toll Brothers Inc., Trex Company Inc., Weyerhaeuser Co., Whirlpool Corp., Worthington Enterprises Inc.

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the second most recent month end: Armstrong World Industries Inc. (\$158.56) and Saint-Gobain (€80.24)

Goldman Sachs has received compensation for investment banking services in the past 12 months: Holcim (SFr77.02) and Saint-Gobain (€80.24)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: Armstrong World Industries Inc. (\$158.56), Holcim (SFr77.02), Kingspan (€81.65), Saint-Gobain (€80.24) and Sika (SFr174.65)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: Armstrong World Industries Inc. (\$158.56), Holcim (SFr77.02), Kingspan (€81.65), Saint-Gobain (€80.24) and Sika (SFr174.65)

Goldman Sachs had a non-investment banking securities-related services client relationship during the past 12 months with: Armstrong World Industries Inc. (\$158.56), Holcim (SFr77.02) and Saint-Gobain (€80.24)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: Armstrong World Industries Inc. (\$158.56), Holcim

(SFr77.02) and Saint-Gobain (€80.24)

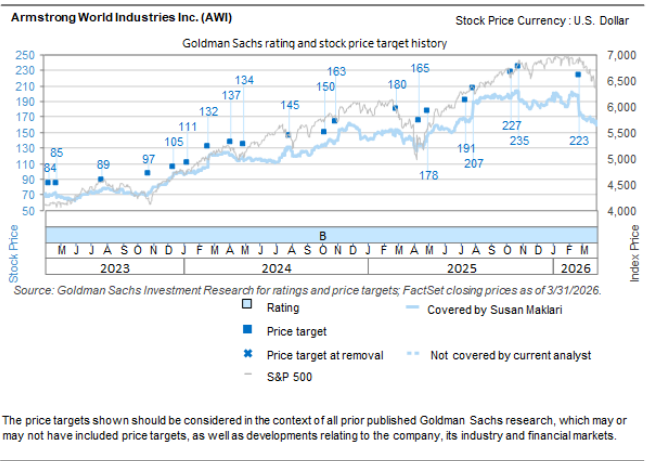
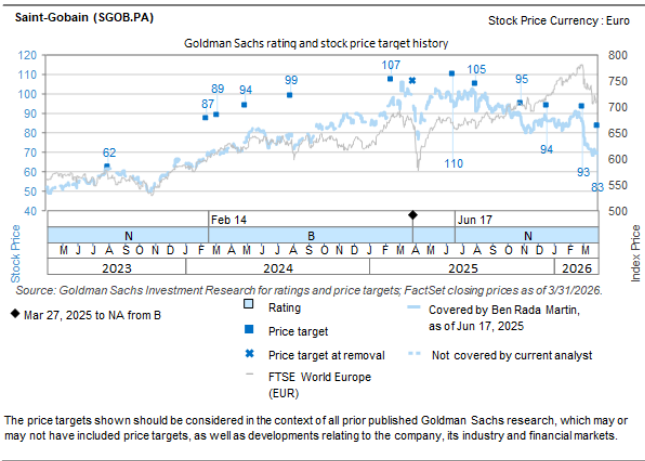
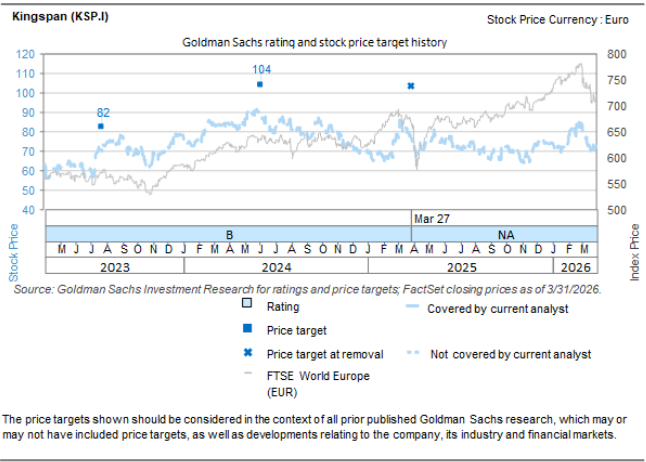
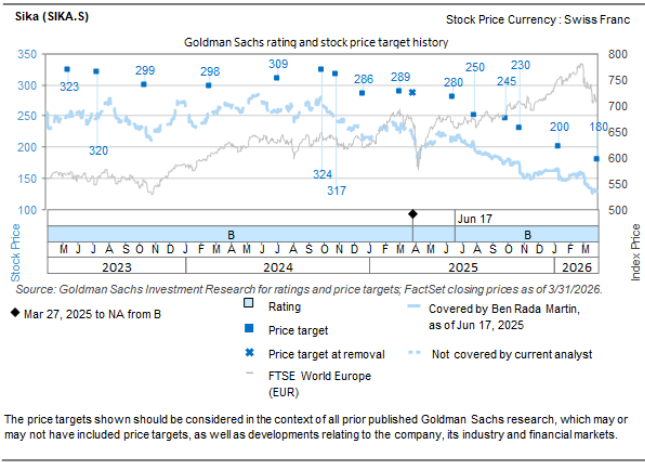
Goldman Sachs makes a market in the securities or derivatives thereof: Armstrong World Industries Inc. (\$158.56)

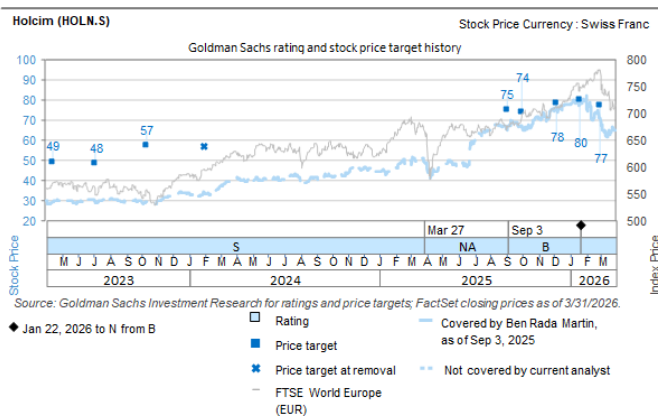
Distribution of ratings/investment banking relationships
Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	60%	45%

As of April 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,074 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)





The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Armstrong World Industries Inc. (AWI)

Date of report	Target price (\$)	Closing price (\$)
28-Apr-26	210.00	169.84
10-Apr-26	206.00	175.48
24-Feb-26	223.00	174.30
29-Oct-25	235.00	191.99
13-Oct-25	227.00	195.85
30-Jul-25	207.00	189.98
15-Jul-25	191.00	164.11
30-Apr-25	178.00	145.02
14-Apr-25	165.00	135.03
27-Feb-25	180.00	151.19
29-Oct-24	163.00	140.00
08-Oct-24	150.00	134.08
30-Jul-24	145.00	132.17
30-Apr-24	134.00	114.88
05-Apr-24	137.00	121.65
20-Feb-24	132.00	117.45
10-Jan-24	111.00	100.45
14-Dec-23	105.00	97.98
24-Oct-23	97.00	73.93
25-Jul-23	89.00	78.03

Sika (SIKA.S)

Date of report	Target price (SFr)	Closing price (SFr)
30-Jun-26	210.00	166.75
14-Apr-26	185.00	148.15
01-Apr-26	180.00	132.20
13-Jan-26	200.00	149.10
27-Oct-25	230.00	168.50
30-Sep-25	245.00	177.10
29-Jul-25	250.00	198.95
17-Jun-25	280.00	215.60
03-Mar-25	289.00	231.80
20-Dec-24	286.00	213.20
28-Oct-24	317.00	251.20
30-Sep-24	324.00	280.10
04-Jul-24	309.00	260.50
20-Feb-24	298.00	255.40
13-Oct-23	299.00	235.60
12-Jul-23	320.00	254.20

Holcim (HOLN.S)

Date of report	Target price (SFr)	Closing price (SFr)
06-Jul-26	88.00	77.02
09-Apr-26	82.00	69.58
01-Mar-26	77.00	70.88
22-Jan-26	80.00	79.90
08-Dec-25	78.00	76.34
02-Oct-25	74.00	67.00
03-Sep-25	75.00	65.08
12-Oct-23	57.00	29.61
06-Jul-23	48.00	28.63

Kingspan (KSP.I)

Date of report	Target price (€)	Closing price (€)
15-Apr-26	80.00	78.55
04-Jun-24	104.00	88.35
25-Jul-23	82.00	71.98

Saint-Gobain (SGOB.PA)

Date of report	Target price (€)	Closing price (€)
23-Apr-26	84.00	78.02
30-Mar-26	83.00	69.36
02-Mar-26	93.00	82.78
19-Dec-25	94.00	86.08
31-Oct-25	95.00	84.06

Date of report	Target price (€)	Closing price (€)
01-Aug-25	105.00	90.98
17-Jun-25	110.00	96.74
14-Feb-25	107.00	96.68
29-Jul-24	99.00	78.00
01-May-24	94.00	74.66
06-Mar-24	89.00	68.14
14-Feb-24	87.00	67.95
03-Aug-23	62.00	60.83

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or fine-tune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.